Chemical, Biological, Radiological and Nuclear (CBRN) Defense Research Institute
Ministry of National Defense, Seoul, Republic of Korea
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Dear Editors,

We submit *Simulants for Chemical Warfare Agent Detection Carry Designations Issued for Other Measurements, or None, and Descriptor Comparison Cannot Supply the Missing Validation* for consideration as a Research Paper.

The manuscript tests a convention this journal recently reviewed. Webb, Coulon and Temple (*J. Hazard. Mater.* **492**, 138021, 2025) state the criterion by which low-toxicity chemical warfare agent simulants are chosen: align hydrophobicity, volatility and solubility with the agent, and the compounds so chosen present minimal toxicity risk. Both halves are testable against existing measurement. We tested them, and we first asked a prior question the criterion presupposes: which document designates these compounds, and for which measurement. The answer to that question is the paper's first result, and it decides how much the descriptor comparison can be asked to carry.

Two of the three incumbent simulants were designated for measurements that are not the one they now support, and the third was never designated at all. We looked for the instrument that designates them. DMMP was identified as a filtration-breakthrough test simulant in a NIOSH respirator programme, in a document that also declines to recommend any simulant. DPGME appears as "a preferred G agent simulant" in a US Army patent, in a training composition carrying a fluorescent dye and justified on ion-mobility detectability. For dichloromethane we found no designating document at all; it is absent from the NIOSH document and from the Man-in-Simulant Test programme. The record separates into three grades, which the manuscript tabulates: a compound named for a stated measurement, a compound recorded only as having been used, and a compound with no document. The clearest designation belongs to none of the three: methyl salicylate is the Army's Man-in-Simulant Test simulant for sulfur mustard and the subject of a published ASTM test method. A simulant is validated against a measurement, not against an agent, and that validation does not travel with the compound into sensor development.

On a panel that contains the compounds the field itself uses as simulants, none of eleven agents has a valid match to an incumbent simulant. For eleven authentic agents we computed RDKit structural and GFN2-xTB electronic descriptors and asked, by exact permutation over every possible designation, whether dimethyl methylphosphonate, dichloromethane and dipropylene glycol monomethyl ether sit closer to the agents than compounds drawn at random. Our primary panel of thirty-five compounds includes five further compounds the sensor, sorbent and decontamination literature uses as simulants (DFP, DCP, DCNP, CEES, methyl salicylate). On it, not one agent is matched, and not one of eighteen tests in three descriptor spaces under six distance statistics reaches nominal significance. The mechanism is not a loss of fidelity. The incumbent compounds sit almost exactly where they sat before: sarin moves from 1.14 to 1.19 typical nearest-neighbour separations from DMMP. What the larger panel supplies is diisopropyl fluorophosphate at 1.07 from sarin, 0.99 from soman and 1.17 from cyclosarin, and 2-chloroethyl ethyl sulfide at 0.57 from sulfur mustard against dichloromethane's 1.67. Resampling the 41 descriptors 1000 times leaves this intact: the incumbent three match a median of zero agents (95% interval zero to three) against five (four to six) for the eight literature-used simulants examined here, and the difference is positive in every replicate. A designation defended by a comparison that excludes the alternatives is not defended.

The twenty-eight-compound panel that omits those five is reported in full, because the contrast is the result. On it, three agents are matched, DMMP supplies all three, and the smallest nominal p is 0.025. How that is corrected decides whether it means anything, so we report five schemes rather than one: under Benjamini-Hochberg across the eighteen tests, across six within each space, at the effective number of independent tests estimated from the correlation of the permutation nulls, or under the Westfall-Young

min-P procedure, which handles the dependence among the eighteen exactly because they share one enumeration of designations, nothing reaches 0.05 (the Westfall-Young value, 0.075, is the closest any correction brings it). Had the statistic been specified in advance, it would survive at $q = 0.046$. We did not specify one. That contrast is the paper's argument made with its own data. The observed value also sits at the 25.8th percentile of 2000 arbitrary three-descriptor subspaces of the same descriptor block.

Taken together, we cannot establish that the incumbent simulants resemble the agents, and we cannot establish that they do not. Three things are open to the analyst: the compound panel, the descriptor space, and the distance statistic. We fixed all three in every combination we could compute, which is eighteen tests on each of three panel and designation configurations, and we report the whole table rather than a cell from it.

One widely used descriptor turned out not to be a coordinate. The Parr electrophilicity index $\omega = \mu^2/2\eta$ is an exact function of the two frontier orbital energies normally reported beside it; across our panel the deviation from the definition reaches at most 1.3×10^{-5} eV on the primary panel and 3.6×10^{-15} eV on the twenty-eight-compound one. A correlation filter does not catch this, because $r(E_{\mathrm{LUMO}}, \omega) = -0.899$. We removed it, which cost the electronic space a dimension it never had, and we state the general form of the point in the Discussion because any descriptor set reporting E_{HOMO}, E_{LUMO}, the gap, μ, η and ω as six quantities has two.

The measured array agrees that the question is unresolved, and corrects us on one point. Twenty-nine-dye colorimetric response for twelve of the compounds, tested by Mantel permutation over compounds rather than over the pairs they generate, gives three nominally significant cells out of twelve that do not agree on a descriptor space, none of which survives correction across the twelve (smallest q = 0.112). Three artifacts of this kind of array analysis are reported alongside, because our own first version contained all three: dye-axis centring fixes the between-compound mean cosine at $-1/(n-1)$, and our observed means clear that floor by only 0.013 to 0.066, so a negative between-compound similarity is arithmetic rather than evidence; six dyes that do not fluoresce under ultraviolet illumination returned one background reading six times, which inflated the two strongest cells until we collapsed them; and the measured agent nearest to DMMP in the ultraviolet channel is tabun, not sarin, although tabun fails our own descriptor matching criterion. We also state there what the three nominal cells cannot carry: all three fall in the plate set that holds no simulant, so they compare agents with agents, while the six cells that bear on the premise are null and the design could only have registered a near-deterministic association. We would rather submit that sentence than a stronger one we cannot support.

The two incumbents differ in safety evidence, not only in hazard. Dichloromethane has a regulator-derived acute inhalation value and it is favourable on that axis: at 41 685 mg m^{-3} it sits near the safe end of the nineteen panel compounds that carry one, though methanol and n-hexane score higher still. The manuscript is explicit that this is the acute axis alone, and that it says nothing about the chronic and carcinogenic classification now displacing dichloromethane from solvent inventories. DMMP has no value at all, because EPA declined to derive an inhalation reference concentration for want of data. Reporting the two as one category of low-toxicity simulant obscures that difference, which matters because DMMP is also the compound against which decontamination efficiency is quantified.

A model trained on the regulatory inventory cannot fill the gap, and the failure is instructive. A random forest on 230 AEGL chemicals reaches $R^2 = 0.458$ with confidence intervals that do not overlap its baseline. Applied to the twenty military agents held out of training, it predicts every one of them safer than measured, by factors of 2.5 to 2600, and certifies fourteen of the twenty as inside its applicability domain. We checked the obvious explanation and it is wrong: thirteen training compounds carry an acetylcholinesterase-inhibiting pharmacophore, identified by substructure and listed in the paper, and one of them is more acutely toxic than sarin. The model fails because it regresses both tails toward the mean, a bias defined in the response space that an input-space applicability domain cannot see.

The two axes are not monotonically associated. Across 207 compounds carrying both measured axes,

Spearman $r = +0.062$ (95% CI -0.077 to $+0.199$). Eight compounds combine top-quartile fidelity with measured acute inhalation safety within an order of magnitude of dichloromethane, all eight clear a structural filter for agent-class motifs, and all eight are more faithful than dichloromethane itself. We propose none of them as a replacement, and the reason is the paper's own argument: a recommendation resting on one panel, one space and two axes would repeat the error we are reporting.

The work sits within the scope of this journal on both sides. The simulant selection literature it tests is published here, and the hazard-screening methods it uses, regulator-derived acute toxicity values bounded by an applicability domain, are the methods this journal applies to other compound classes.

On data integrity, and on this manuscript's own history. This analysis was audited four times before submission and each audit changed the numbers, so we state what happened rather than present the final version as though it had come out that way.

The first audit found that the numeric parser had split thousands separators in the source table, so that "12,000 ppm" was read as 12. The second, of chemical identity rather than of numbers, found two records whose stored structure belonged to a different compound; both are excluded. The third found four further faults in how the regulatory table was read, all of one kind: treating a printed table as though its cells always held plain numbers. Footnote lines beginning *AEGL 3 - 10 min = ...* were parsed as though they were table rows, which turned duration labels into values and gave propane and butane ten-minute AEGL-3 values of 10 ppm against the true 33,000 and 77,000 ppm. Nine compounds whose ten-minute cell reads *see below*, the agency's marker for a value above half the lower explosive limit, had no value collected at all, among them methanol, acetone, benzene and n-hexane. Rows whose longer durations read *NR* were discarded whole, which silently dropped three arsenical vesicants and BZ from the held-out agent set. And one row broken across two lines by the document's text layer was lost.

The fourth audit was of our own statistics rather than of the data, and it is the reason the paper now says what it says. It found the functional dependence in the electronic descriptor set, the pairwise-distance pseudoreplication in the array analysis, the algebraic floor created by centring, the absence of any multiplicity correction across eighteen tests we had in fact computed, and the fact that we had never run the panel that includes the other compounds the literature treats as simulants. An earlier version of this manuscript reported $p = 0.028$ as a result and built a thesis on it. None of those five problems is visible in that version, and together they remove the thesis. We have rewritten the paper around what the data support, and we report the earlier reading alongside it rather than deleting it, because how a nominal result of this kind arises is part of what a reader of this literature needs.

The extraction is now cell-based and refuses any row that does not resolve into five duration cells, and it is validated over the whole table by a second reader that shares neither the text-extraction engine nor the algorithm, reading values by character column position rather than by token order. The two readers agree on all 259 entries, on all 257 of those that resolved to a structure and on all nineteen panel values.

Dual-use. The optimisation direction is fixed throughout and stated in the manuscript: detection-relevant fidelity is maximised while phosphorus electrophilicity, acetylcholinesterase reactivity, alkylating capacity and predicted mammalian toxicity are minimised. The structural filter excludes candidates and never orders them. The one ordering the paper publishes ranks compounds that have already cleared that filter by resemblance to the agents, and every compound in it is an industrial chemical in commerce carrying a published regulatory exposure value. We did not extend the toxicity model to the organophosphorus class, because a predictor of mammalian lethality evaluated over structures not yet made, inside a loop that proposes them, is the configuration this work is built to avoid. We cite the existing literature on organophosphorus lethality modelling, including our own, rather than implying that none exists.

The acute toxicity axis, the random-forest interpolation of it, the decoupling test between fidelity and toxicity, and the phosphorus-local reactivity analysis are in the Supplementary Material. They stand on their own data and are summarised where the main text needs them; we placed them there because the

argument of the paper turns on the composition of the comparison panel and does not depend on them.

The manuscript is original, is not under consideration elsewhere, and has been approved by all six authors. The authors declare no competing financial interest, and the work received no specific funding. Analysis code, descriptor tables, parsed toxicity values with their sources and all figure scripts are deposited in the public repository cited in the manuscript and also accompany the submission. The exact permutation tests are exhaustive and the remaining analyses are deterministic given the stated seeds.

Thank you for considering this work.

Yours sincerely,

Doo-Hee Lee and Yoon Jeong Jang, corresponding authors
on behalf of Ku Kang, Jin Yoo, Jeongyun Kim, Miae Kim, Doo-Hee Lee
and Yoon Jeong Jang
dooheelee@korea.kr yjjangchem@korea.kr